Soy food intake after diagnosis of breast cancer and survival: an in-depth analysis of combined evidence from cohort studies of US and Chinese women

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Background: Soy isoflavones have antiestrogenic and anticancer properties but also possess estrogen-like properties, which has raised concern about soy food consumption among breast cancer survivors. Objective: We prospectively evaluated the association between postdiagnosis soy food consumption and breast cancer outcomes among US and Chinese women by using data from the After Breast Cancer Pooling Project. Design: The analysis included 9514 breast cancer survivors with a diagnosis of invasive breast cancer between 1991 and 2006 from 2 US cohorts and 1 Chinese cohort. Soy isoflavone intake (mg/d) was measured with validated food-frequency questionnaires. HRs and 95% CIs were estimated by using delayed-entry Cox regression models, adjusted for sociodemographic, clinical, and lifestyle factors. Results: After a mean follow-up of 7.4 y, we identified 1171 total deaths (881 from breast cancer) and 1348 recurrences. Despite large differences in soy isoflavone intake by country, isoflavone consumption was inversely associated with recurrence among both US and Chinese women, regardless of whether data were analyzed separately by country or combined. No heterogeneity was observed. In the pooled analysis, consumption of ≥10 mg isoflavones/d was associated with a nonsignificant reduced risk of all-cause (HR: 0.87; 95% CI: 0.70, 1.10) and breast cancer–specific (HR: 0.83; 95% CI: 0.64, 1.07) mortality and a statistically significant reduced risk of recurrence (HR: 0.75; 95% CI: 0.61, 0.92). Conclusion: In this large study of combined data on US and Chinese women, postdiagnosis soy food consumption of ≥10 mg isoflavones/d was associated with a nonsignificant reduced risk of breast cancer–specific mortality and a statistically significant reduced risk of recurrence. One of the studies included in the After Breast Cancer Pooling Project, the Women's Healthy Eating & Living Study, was registered at clinicaltrials.gov as NCT00003787.